

Exam 1

1:10 to 3:05 PM

NAME: _____

Question 1 True, False, or Uncertain (20 points). Answer true, false, or uncertain for the following statements and justify your answer with a brief explanation.

- 1) (5 points) If the Japanese CPI is currently at 108 and the US CPI is at 104, then the Japanese rate of inflation is higher than the US rate of inflation.

- 2) (5 points) An increase of one unit in government spending leads to an increase of one unit in equilibrium output.

- 3) (5 points) The term *investment*, as used by macroeconomists, refers to the purchase of bonds and shares of stock.

- 4) (5 points) If investment is exogenous, *IS* curve is horizontal on $Y - i$ plane.

To be continued to the next page ...

- 1) (5 points) Provide a definition of the natural rate of unemployment.
- 2) (5 points) Provide a definition of the property of constant returns to scale.
- 3) (10 points) Describe briefly three time frames in macroeconomics: short run, medium run, and long run.
- 4) (10 points) Show that the growth rate of real output can be approximated by log difference:

$$\text{If } g = \frac{Y_t - Y_{t-1}}{Y_{t-1}},$$

$$g \approx \log Y_t - \log Y_{t-1}$$

Question 3 (30 points). Suppose the current economy stays at the medium-run equilibrium and is characterized by the following behavioral equations:

$$C = 200 + 0.5Y_D$$

$$I = 100 + 0.1Y - 2,000r$$

$$G = 400$$

$$T = 400$$

$$\bar{r} = 0.05$$

$$\frac{M^d}{P} = 2Y - 10,000r$$

Aggregate Production Function : $Y = N$

Fixed Labor Force : $\bar{L} = N + U$

WS Relation : $\frac{W}{P} = (1 - u)$

$$\text{PS Relation : } \frac{W}{P} = \frac{1}{1+m}$$

- 1) (3 points) Explain the meaning of the IS curve and derive IS relation.

- 2) (6 points) Solve for the equilibrium values of output Y_n , consumption C_n , investment I_n , and real money supply $\frac{M^s}{P}$. Draw the equilibrium of this economy using an $IS - LM - PC$ diagram.

Now suppose that government increases G permanently by 200.

- 3) (5 points) Solve for the new equilibrium values of output Y' , consumption C' , and investment I' in the short run. Draw the effects of expansionary fiscal policy on an $IS - LM - PC$ diagram.

- 4) (4 points) To maintain the existing medium-run equilibrium of output, in what direction and how much must the central bank change r ? What is the level of real money supply $\left(\frac{M^s}{P}\right)'$ with the changed interest rate at the existing medium-run level of output? Draw the effects of monetary policy on an $IS - LM - PC$ diagram.

- 5) (4 points) Now suppose that the central bank wants to adhere to the previous policy rate of real rate $\bar{r} = 0.05$. In this case, to maintain the existing medium-run equilibrium of output, the government can increase T permanently instead of implementing monetary policy. How much must the government increase T ? Draw the effects of change in T on an $IS - LM - PC$ diagram. (*Hint: consider the multiplier effect in this economy.*)

Now return to the initial situation in which $G = 400$, $T = 400$, and $\bar{r} = 0.05$.

- 6) (4 points) Suppose that the markup of goods prices m is 25%. Solve for the real wage, labor force, and natural rate of unemployment at the current medium-run equilibrium. (*Hint: Note that $Y_n = N_n$.*)

- 7) (4 points) Now suppose that the mark up increases to 100% because of an increase in oil prices. Solve for new real wage, new natural rate of unemployment, and new natural level of output. Draw the effects of an increase in markup on an $IS - LM - PC$ diagram.

- 4) (6 points) What is the definition of the golden rule of saving rate? Solve for the steady-state level of consumption per effective worker when the golden rule is satisfied. Compare with the level of consumption per effective worker obtained in 3).